

DAVID OPREA

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Education

Purdue University
B.S. in Computer Science
GPA: 4.0/4.0

West Lafayette, IN
Expected May 2028

Skills

- **Languages:** Python, Java, C++, Julia, JavaScript, HTML/CSS
- **Frameworks & Tools:** PyTorch, NumPy, Pandas, Matplotlib, React, Node.js, Flask, Git, GitHub, Arduino
- **Topics:** Algorithms, Data Structures, Machine Learning, Quantitative Analysis, Circuits, Web Development

Projects

- **ReclarifAI** — *AI-Powered Learning Application*
 - Engineered a full-stack web application during a 24-hour hackathon to help users study more effectively by preventing rote memorization of definitions, **improving upon the gameplay of Quizlet Match**.
 - Designed and implemented a responsive front-end using **React** and a **Flask** back-end to manage user data and Open AI API calls, generate unique, yet similar definitions for flashcard terms in real-time.
- **Pioneer Research** — *Algorithms Researcher*
 - Developed a novel heuristic for the Minimum Set Cover problem using wave function collapse, reducing solution size by **1.18%** over both a Greedy algorithm and a Big Greedy Algorithm.
 - Optimized the heuristic to be over **500x faster** than the leading Tabu Search method, enabling analysis of larger datasets.
- **Lumiere Education** — *A.I. Researcher*
 - Tested a new method (BVM) to judge image state changes via language embeddings, achieving **0.91% greater accuracy** in discerning noun-adjective pairs than Naive Bayes and better accuracies against other leading methods.

Activities & Leadership

- **ML@Purdue** — *Machine Learning Researcher (VGC-RL Team)* Sept 2025 - Present
 - Architecting a Reinforcement Learning agent for competitive Pokémon VGC in PyTorch with a policy-driven Monte Carlo Tree Search (MCTS) to achieve a >90% win rate against baseline models.
- **Boiler Quant** — *Quantitative Analyst* Sept 2025 - Present
 - Creating a social stock exchange platform by implementing a full-featured order book, establishing exchange market mechanics, and designing a liquidity rewards program to incentivize market makers.
- **Purdue Stack** — *Full Stack Developer* Sept 2025 - Present
 - Developing a standalone desktop GUI for seismologists using React and Electron for interactive earthquake maps, waveform plots, and spectrograms to analyze and export complex scientific data.
- **Science Olympiad** — *Event Leader & Officer* 2022 - 2025
 - Built a redox probe using Arduino with under **5% error**, securing **1st place** at the Princeton Invitational.

Awards

- **USACO Platinum Division** (Top 200 U.S. High School Competitors)
- **National Cyber Scholarship Finalist** (Top 1000 U.S. High School Students)
- **Science Olympiad:** 30+ medals across invitational, regional, and state competitions